Fusion Constructive Realism (FCR)

Complete Mathematical Specification, State-Space Dynamics, and Non-Uniform Kinetic Formalism

Version 2.0 Theory Specification and Architecture Document

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Abstract

This document provides the complete, self-contained, and mathematically unified specification of Fusion Constructive Realism (FCR v2.0). FCR is a non-local, constructive, background-independent relational network framework wherein spacetime geometry, quantum mechanical interference, and particle topologies emerge from discrete relational correlation state spaces. This revised specification resolves all foundational ambiguities present in prior summaries: (1) we establish an explicit architectural distinction between internal correlation graph micro-paths and macro-trajectories in the branching world-state transition space $\mathcal{M}$ under poset time evolution $T \to T'$; (2) we eliminate metric collapse and negative distance pathologies via a non-linear asymptotic update law; (3) we derive physically grounded non-uniform mutation weights $W(m)$ from thermodynamic and spatial locality principles; (4) we resolve phase venting zero-sum singularities via global 3-cycle loop trace gauge fixing $\mathrm{Tr}(C^3)$ with phase recoil conservation; and (5) we specify an epistemologically aligned canonical reduction pipeline for state space deduplication.

Contents

1 Foundational Ontology and State Representation

Fusion Constructive Realism rejects pre-existing continuous spacetime manifolds, absolute coordinates, and local realism. Physical reality is defined entirely as a discrete relational state $Q = (P, \mu, C)$ evolving across poset step transitions $T \to T'$.

1.1 State Triad Definition

Definition 1.1 (World State Q)**.** *A world state $Q \in \mathcal{M}_n$ is represented by a triad (P, μ, C):*

1. ***Base Set (P):*** *A finite set of n discrete node identifiers $P = \{0, 1, \ldots, n-1\}$, representing elementary relational points.*

2. ***Complex Correlation Field (C):*** *An $n \times n$ Hermitian-like matrix $C(i, j) \in \mathbb{C}$ encoding complex phase and correlation strength between nodes $i, j \in P$, subject to:*

$$|C(i, j)| \leq 1.0, \quad C(i, i) = 1.0 + 0.0i, \quad C(i, j) = C(j, i)^* \tag{1}$$

3. ***Primitive Metric (μ):*** *A spatial distance matrix $\mu(i, j) \in \mathbb{R}^+$ satisfying:*

$$\mu(i, j) = \begin{cases} 1.0 & \textit{(baseline rest metric), if } C(i, j) \neq 0 \\ \text{ShortestPath}_\mu(i \to j) & \textit{if } C(i, j) = 0 \\ \infty & \textit{if disjoint / disconnected} \end{cases} \tag{2}$$

with self-distance $\mu(i, i) = 0.0$.

1.2 Node Capacity Bound Axiom

Axiom 1.1 (Node Capacity Limit)**.** *Physical stability requires that no single node can support unbounded external correlations. For every node $p \in P$, the external correlation capacity sum $S(p)$ must satisfy:*

$$S(p) = \sum_{q \neq p} |C(p, q)|^2 \leq 1.0 \tag{3}$$

2 Dual-Graph Formalism: Internal vs. Macro State-Space

A critical structural clarification in FCR 2.0 is the strict separation between internal graph states and the global state-space transition graph.

Definition 2.1 (Internal Correlation Graph Q)**.** *The internal graph $Q = (P, \mu, C)$ represents a single micro-state of the relational universe. Internal paths $\gamma = (e_1, e_2, \ldots, e_k)$ connect nodes within P.*

Definition 2.2 (Macro World-State Graph $\mathcal{M}$)**.** *The macro state-space graph $\mathcal{M} = (\mathcal{V}_\mathcal{M}, \mathcal{E}_\mathcal{M})$ is a directed, branching transition graph where:*

- *Vertices $[Q] \in \mathcal{V}_\mathcal{M}$ are canonical equivalence classes of world-states under gauge rotation and node permutation isomorphism.*

- *Directed arrows $Q_T \xrightarrow{m} Q'_{T'} \in \mathcal{E}_\mathcal{M}$ represent elementary local operational mutations $m \in \Sigma$ over poset time $T \to T'$.*

3 Targeted Operational Mutation Taxonomy (Σ) and Physical Kinetics

Evolution across poset steps $T \to T'$ occurs via operational mutations from the alphabet $\Sigma = \Sigma_{\text{lawful}} \cup \Sigma_{\text{contingent}}$. Rather than applying mutations globally and deterministically, an isolated world-state Q_T branches into all valid single local mutations $m(Q_T) = Q'_{T'}$.

3.1 Lawful Operational Mutations ($\Sigma_{\text{lawful}} = \{t, d, \mu, v\}$)

Lawful mutations govern the internal, closed-system unitary kinetics of an isolated relational world Q.

3.1.1 Triangulation (t): Structural Closure

Triangulation acts as a topological closure operation on open 2-paths $i \to x \to j$ where $C(i, j) = 0$:

$$C(i, j) \leftarrow C(i, x) \cdot C(x, j) \quad \text{iff } C(i, j) = 0 \tag{4}$$

Physical Mutation Weight (W_t): To enforce spatial locality and prevent instantaneous non-local signaling, triangulation carries a transition weight that decays exponentially with spatial distance:

$$W_t = e^{-\mu(i,j)} \tag{5}$$

3.1.2 Correlation Decay (d)

Correlation magnitudes dissipate across spatial distance:

$$C_{T'}(i, j) = C_T(i, j) \cdot e^{-\gamma_d \mu_T(i,j)} \tag{6}$$

where $\gamma_d > 0$ is the decay constant. If $|C(i, j)| < \epsilon_{\text{sever}} = 10^{-6}$, the edge is severed ($C(i, j) \to 0$). **Physical Mutation Weight (W_d):** Dissipative decay across large metric separations is favored thermodynamically:

$$W_d = e^{+\mu_T(i,j)} \tag{7}$$

3.1.3 Asymptotic Metric Evolution (μ)

To resolve the negative distance pathology ($\mu \leq 0$) of linear updates, metric distance contracts under dense correlation and relaxes toward unit rest distance $\mu = 1.0$ according to an **asymptotic non-linear update law**:

$$\mu_{T'}(i, j) = \mu_T(i, j) \cdot e^{-\alpha|C_T(i,j)|} + \beta(1.0 - \mu_T(i, j)) \tag{8}$$

where $\alpha > 0$ is the metric contraction coupling and $\beta \in (0, 1)$ is spatial elasticity recovery.

Proposition 3.1 (Metric Positivity Guarantee)**.** *Rewriting the update as $\mu_{T'} = \mu_T(e^{-\alpha|C|} - \beta) + \beta$, since $e^{-\alpha|C|} > 0$ and $\beta \in (0, 1)$, $\mu_{T'} > 0$ strictly holds for all $\mu_T \in (0, \infty)$ across all finite poset transitions $T \to T'$.*

Physical Mutation Weight (W_μ): Metric adaptation scales directly with correlation magnitude:

$$W_\mu = |C_T(i, j)| \tag{9}$$

3.1.4 Phase Venting ($\mathcal{M}_{1/2}$) and Topological Fission (v)

When a mutation causes a node p to exceed unit capacity ($S(p) > 1.0$), phase venting acts as an **instantaneous virtual transaction** ($W_v = 1.0$, priority execution before advancing poset step $T \to T'$).

Global 3-Cycle Loop Trace Gauge and Phase Recoil Node p sheds excess capacity magnitude $\Delta S = S(p) - 1.0$. To eliminate zero-sum indeterminate singularities ($\arg(0)$) when neighboring correlations cancel ($\left|\sum_{q\neq p} C(p,q)\right| \leq \epsilon_{\text{cancel}} = 10^{-6}$), ϕ_{vent} is anchored to the global 3-cycle loop trace phase of the cosmic vacuum:

$$\Phi_{\text{global}} = \arg\left(\text{Tr}(C^3)\right) = \arg\left(\sum_{i,j,k\in P} C(i,j)C(j,k)C(k,i)\right) \tag{10}$$

The venting phase is determined deterministically by:

$$\phi_{\text{vent}} = \begin{cases} \arg\left(\sum_{q\neq p} C(p,q)\right) & \text{if } \left|\sum_{q\neq p} C(p,q)\right| > \epsilon_{\text{cancel}} \\ \Phi_{\text{global}} & \text{otherwise} \end{cases} \tag{11}$$

Phase Recoil Conservation: To conserve net phase momentum when emitting against Φ_{global}, node p's active edges absorb an equal and opposite phase recoil: $C(p,q) \leftarrow C(p,q) \cdot e^{-i\phi_{\text{vent}}}$.

Coherent Amplitude Scaling & Fission All active edge correlations connected to overloaded node p are projected back to unit capacity:

$$C(p,q) \leftarrow \frac{C(p,q)}{\sqrt{S(p)}} \implies \sum_{q\neq p}\left|\frac{C(p,q)}{\sqrt{S(p)}}\right|^2 = 1.0 \tag{12}$$

Subgraphs disconnected by nilpotent link severing ($|C| < \epsilon_{\text{sever}} = 10^{-6}$) are ejected as radiation remnants ($\mathcal{M}_2$ chains).

3.2 Contingent Measurement Operations ($\Sigma_{\text{contingent}} = \{f\}$)

Unlike lawful kinetics, Fusion (f) is a non-deterministic, contingent measurement event representing an open interaction between two previously disconnected relational sub-universes Q_A and Q_B. Fusion cannot be retroactively predicted by the closed laws of Q_A or Q_B alone, but is confirmed *ex post facto* by boundary observation.

3.2.1 Multi-Node Boundary Geodesics

Two disjoint connected components $Q_A = (P_A, \mu_A, C_A)$ and $Q_B = (P_B, \mu_B, C_B)$ fuse along boundary subsets $\partial P_A \subseteq P_A$ and $\partial P_B \subseteq P_B$ subject to boundary metric equivalence. The initial cross-boundary metric between node $i \in P_A$ and $j \in P_B$ is defined by the shortest path across the bridge edge set E_{bridge}:

$$\mu(i,j) = \min_{(a,b)\in E_{\text{bridge}}} \left(\mu_A(i,a) + \mu_{\text{bridge}}(a,b) + \mu_B(b,j)\right) \tag{13}$$

where baseline bridge distance is $\mu_{\text{bridge}}(a,b) = 1.0$.

4 State-Space Trajectory Integrals and Display Postulate

4.1 Trajectory Amplitudes in $\mathcal{M}$

A macro-trajectory Γ in state-space $\mathcal{M}$ from initial state Q_0 to target state Q_k is a sequence of local mutations:

$$\Gamma = \left(Q_0 \xrightarrow{m_1} Q_1 \xrightarrow{m_2} \dots \xrightarrow{m_k} Q_k\right) \tag{14}$$

The total transition weight $A(\Gamma)$ of trajectory Γ is given purely by the real product of physical mutation kinetics:

$$A(\Gamma) = \prod_{i=1}^{k} W(m_i) \tag{15}$$

Quantum phase interference arises naturally from complex correlation fields $C(i,j)$ embedded within convergent target states $[Q]$.

4.2 Isomorphic Trajectory Merging and Display Postulate

When multiple distinct trajectories $\Gamma_1, \Gamma_2 \in \mathcal{M}$ land on isomorphic states, canonical deduplication collapses them to a single vertex $[Q] \in \mathcal{V}_{\mathcal{M}}$ without modifying internal edge correlations $C(i,j)$. The total trajectory amplitude sums additively: $A_{\text{total}}([Q]) = \sum_{\Gamma \to [Q]} A(\Gamma)$.

The probability $P([Q])$ of displaying canonical state class $[Q]$ at step T_k is governed by the empirical Born-rule display postulate applied at the observational boundary:

$$P([Q]) = \frac{\left|\sum_{\Gamma \to [Q]} A(\Gamma)\right|^2}{\sum_{[Q'] \in \mathcal{V}_{\mathcal{M}}(T_k)} \left|\sum_{\Gamma' \to [Q']} A(\Gamma')\right|^2} \tag{16}$$

5 Attractor Hierarchy and Wave-Packet Particle Motion

5.1 Mechanism of Class-2 Stable Attractors

Definition 5.1 (Class-2 Attractor Shell). *A Class-2 Attractor is a closed topological shell (such as the $\mathcal{M}_3$ triangle or $\mathcal{M}_6$ octahedron) forming a stable limit cycle where metric contraction ($\mu \to 0$) exactly balances correlation decay ($e^{-\gamma_d \mu} \to 1.0$). At equilibrium ($\mu^*, |C^*|$):*

$$|C_{T'}| = |C_T| > 0, \quad \mu_{T'} = \mu_T > 0 \tag{17}$$

Phase rotates indefinitely around closed loops without dissipation or capacity blowup, forming persistent elementary particles.

Definition 5.2 (Attractor Hierarchy Spectrum).

- ***Class-1 (Static / Severed):*** *Transient tree chains ($\mathcal{M}_2$) that dissipate rapidly into radiation remnants.*
- ***Class-2 (Cyclic / Particles):*** *Stable, compact topological shells forming fundamental matter states.*
- ***Class-3 (Quasi-Periodic / Toroidal):*** *Multi-loop nested topologies exhibiting complex internal quantum spin numbers.*
- ***Class-4 (Adaptive / Chaotic Networks):*** *Open-ended macroscopic relational networks capable of macro-scale classical spacetime collapse.*

5.2 Particle Wave-Packet Motion

In FCR, particles do not move through a background coordinate space. Particle motion is modeled as a **wave packet transfer across valence nodes in** $\mathcal{M}$: correlation density $|C(i,x)|$ decays on one side while triangulating (t) and contracting (μ) on another, causing the center of node valence of the Class-2 attractor shell to shift across the state network.

6 Permutation-Invariant Canonical Reduction Pipeline

To ensure state-space deduplication, candidate states pass through a 3-step reduction algorithm:

1. **Global Loop Phase Gauge-Fixing:** Rotate all edge phases relative to the global 3-cycle loop phase $\Phi_{\text{global}} = \arg(\text{Tr}(C^3))$:

$$C_{\text{gauge}}(a,b) = C(a,b) \cdot e^{-i\Phi_{\text{global}}} \implies \arg(\text{Tr}(C^3_{\text{gauge}})) = 0 \tag{18}$$

2. **Numerical Binning (Computational Practicality):** Round floating-point values for $|z|$, $\arg(z)$, and μ to fixed precision $\epsilon_{\text{bin}} = 10^{-3}$. This step is a pragmatic software choice for classical hardware simulation to eliminate numerical float drift.

3. **Canonical Graph Hashing (Classical Processor Artifact):** Pass the binned matrix to the Weisfeiler-Lehman / VF2 graph hash algorithm, producing a unique deterministic string hash. On quantum computational substrates, physical isomorphism collapse is resolved natively by relational nature without explicit graph hashing pipelines.

7 Comparison and Resolution Matrix (FCR v1.0 vs. v2.0)

Theoretical Dimension	**Summary Document (v1.0)**	**FCR Formal Specification (v2.0)**
Time Formalism	Linear time step $T+1$	Causal poset transition $T \to T'$
State Space Domain	Conflated internal paths & $\mathcal{M}$ trajectories	Distinct Q-graph vs. $\mathcal{M}$ state-space graph
Metric Stability	Linear update ($\mu \to$ negative)	Asymptotic update $\mu_{T'} = \mu_T e^{-\alpha\|C\|} + \beta(1 -$
Mutation Taxonomy	Implicit global update	Lawful Σ_{lawful} vs. Contingent $\Sigma_{\text{contingent}} = \{$
Mutation Kinetics	Uniform / Unspecified	Non-uniform weights $(W_t = e^{-\mu}, W_d = e^{\mu}, W$
Venting Singularity	Undefined $\arg(0)$ if sum cancels	Global loop trace gauge Φ_{global} with phase re
Trajectory Amplitudes	Artificial complex phase $e^{i\Theta}$	Pure real weights $A(\Gamma) = \prod W(m)$; internal
Gauge-Fixing Basis	First non-zero edge (index-dependent)	Topological trace loop phase $\arg(\text{Tr}(C^3))$
Canonical Reduction	Hardcoded algorithm	Recognized as classical computation artifact

Table 1: Key Mathematical and Algorithmic Improvements in FCR Version 2.0.

8 Conclusion

Fusion Constructive Realism 2.0 establishes a completely rigorous, self-contained, and singularity-free mathematical theory. By resolving metric update instabilities, defining non-uniform mutation kinetics, anchoring phase venting to global loop invariants with phase recoil, classifying lawful versus contingent operations, and clarifying the state-space trajectory display postulate, FCR 2.0 provides an unambiguous basis for computational simulation and fundamental physics modeling.